

SMAYAN A. KHANNA

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EDUCATION

University of California San Diego (UCSD), San Diego, CA

September 2026 – Expected June 2031

- Doctor of Philosophy (PhD) in Data Science

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The University of Chicago, Chicago, IL

September 2025 – Expected June 2026

- Master of Science (MS) in Computer Science
- **Advanced Scholars Program, GPA: 4.0/4.0**

The University of Chicago, Chicago, IL

September 2021 – June 2025

- Bachelor of Science (BS) in Physics with a Computer Science Minor
- **Cum Laude, Dean's List (21-22, 23-24 & 24-25), GPA: 3.872/4.0, Major GPA: 3.92/4.0**

Relevant Coursework: Machine Learning, NLP, Deep Active Learning, Algorithms Analysis, Probability, Data structures, Discrete Mathematics, Abstract Linear Algebra, Differential Equations, Quantum Mechanics, Soft Matter Physics

PAPERS & PUBLICATIONS

Khanna, S., Gökmen, D.E., Kondor R., & Vitelli, V. (2025). Graph Contrastive Learning vs Untrained Baselines: The Role of Dataset Size. *Preprint*. <https://arxiv.org/abs/2509.01541>.

Jones, M., **Khanna, S.**, & Ferguson, A. (2025). FlowBack: Generalized Flow-Matching Approach for Biomolecular Backmapping. Published to *Journal of Chemical Information and Modeling (JCIM)*, ACS Publications. <https://pubs.acs.org/doi/10.1021/acs.jcim.4c02046>.

Jones, M., **Khanna, S.**, & Ferguson, A. (2024). FlowBack: A Flow-matching Approach for Generative Backmapping of Macromolecules. *ICML 2024 Workshop on Machine Learning for Life Sciences (ML4LMS)*. [Oral Presentation](#). Vienna, Austria.

Ashwood, B., Jones, M. S., Radakovic, A., **Khanna, S.**, Lee, Y., Sachleben, J. R., Szostak, J. W., Ferguson, A. L., & Tokmakoff, A. (2023). Thermodynamics and kinetics of DNA and RNA dinucleotide hybridization to gaps and overhangs. *Biophysical Journal*, 122(16). 3323–3339. <https://doi.org/10.1016/j.bpj.2023.07.009>.

PRESENTATIONS

FlowBack: A Flow-matching Approach for Generative Backmapping of Biomolecules. Poster presented at the *UChicago Undergraduate Research Symposium*, Chicago, IL. 2024.

Thermodynamics and kinetics of DNA and RNA dinucleotide hybridization to gaps and overhangs. Poster presented at the *Midstates Undergraduate Research Symposia*, Chicago, IL. 2023.

RESEARCH EXPERIENCE

University of Chicago - James Franck Institute, Vitelli Group, Chicago, IL

December 2024 – Present

Research Assistant (including full-time summer)

- Researching deep-learning techniques for representation learning in biological and physical systems
- Developed a contrastive algorithm, using Graph Neural Networks (GNNs) and statistical physics techniques to optimally coarse-grain inhomogeneous graph data
- Led a first-author research project analyzing the effectiveness and scaling behavior of Graph Contrastive Learning (GCL) for unsupervised graph classification

Massachusetts Institute of Technology (MIT), Dunkel Group, Cambridge, MA

June 2023 – August 2023

Undergraduate Research Intern

- Conducted research with Prof. Jörn Dunkel and the Flavell Lab on the mathematical brain-behavior link in *C. elegans*
- Developed a Julia pipeline which facilitated the analysis and visualization of neural trace activity datasets
- Applied PCA, graph theory, and compressed sensing in Julia for dimensionality reduction of neural circuits
- Identified a superior neural basis for the signal reconstruction of timeseries data using Matching Pursuit, outperforming randomly sampled bases by ~20% reconstruction accuracy

Pritzker School of Molecular Engineering, Ferguson Lab, Chicago, IL

June 2022 – June 2024

Undergraduate Research Assistant (Sep 2022 – Jun 2024)

- Co-developed a generative model using Conditional Flow Matching and Equivariant Graph Neural Networks

- (EGNNs) to back-map protein and DNA traces from coarse-grained simulations
- Trained and fine-tuned the model on DNA-protein structures, optimizing inference time by evaluating different ODE solvers and achieving over 99% reconstruction accuracy, outperforming previous models by 3%

Undergraduate Research Intern (May 2022 – Aug 2022)

- Conducted research with the Tokmakoff Group and Prof. Andrew Ferguson on T-jump spectroscopy by generating all-atom molecular dynamics simulations of DNA
- Streamlined the creation of nucleotide topologies, replica exchange simulations, and cluster jobs through automation scripts

PROJECTS

Deep Active Learning with Graph Neural Networks

April 2025 – June 2025

- Developed a Deep Active Learning (DAL) pipeline for scalable graph classification using Graph Neural Networks (GNNs) in Pytorch Geometric

Investigating Linguistic Complexity in LLMs

January 2025 – March 2025

- Developed a novel Activation Patching & Vector Patching method using Transformer Lens to probe Syntactic & Linguistic complexity in GPT-2
- Selected as Best Class Project (faculty-rated) for CMSC 25700: *Natural Language Processing*

Bot Checkers

January 2023 – May 2023

- Developed the GUI for an N-boards checkers game with integrated bot logic using Pygame

TEACHING EXPERIENCE

University of Chicago, Chicago, IL

March 2025 – Present

Teaching Assistant (TA): DATA 35900 - Responsible Use of Data and Algorithms

- Sole TA for a graduate-level course on Differential Privacy & Algorithmic Fairness
- Designed and graded assignments, supported course organization and student learning

Instructional Grader: CMSC 25400 - Machine Learning

- Assisted with grading in the core Machine Learning course at UChicago

The Science Biome, Mumbai, India

May 2016 - Present

Founder

- Researched, wrote, and animated 31 science videos on physics, math, and cosmology
- Amassed 2K subscribers and 90K+ views on the YouTube channel ([link](#))
- Recognized by Cornell Prof. Steven Strogatz for a video project on the applications of Differential Equations to HIV research and treatment ([tweet](#), [video](#))
- Interviewed by the Mumbai Mirror ([link](#)), and the national newspaper, The Times of India ([link](#))

AWARDS

2023-2024 Quad Undergraduate Research Scholarship, *University of Chicago*. \$5000.

2023 Jeff Metcalf Internship Fellowship. *University of Chicago*. \$4000.

2021 Runner-up, Strogatz Prize for Math Communication. *National Museum of Mathematics (MoMath)*, USA. \$314.159.

2020 Breakthrough Junior Challenge Semifinalist, *Breakthrough Prize Organization, Khan Academy & National Geographic*.

PROGRAMMING SKILLS

Python (Advanced): Pytorch, Pytorch Geometric, HuggingFace Transformers, Scikit Learn, TensorFlow, Numpy

Julia (Intermediate): DifferentialEquations.jl, MultivariateStats.jl, Graphs.jl

Others (Beginner): C, Haskell

Tools: Git, CUDA, SLURM